

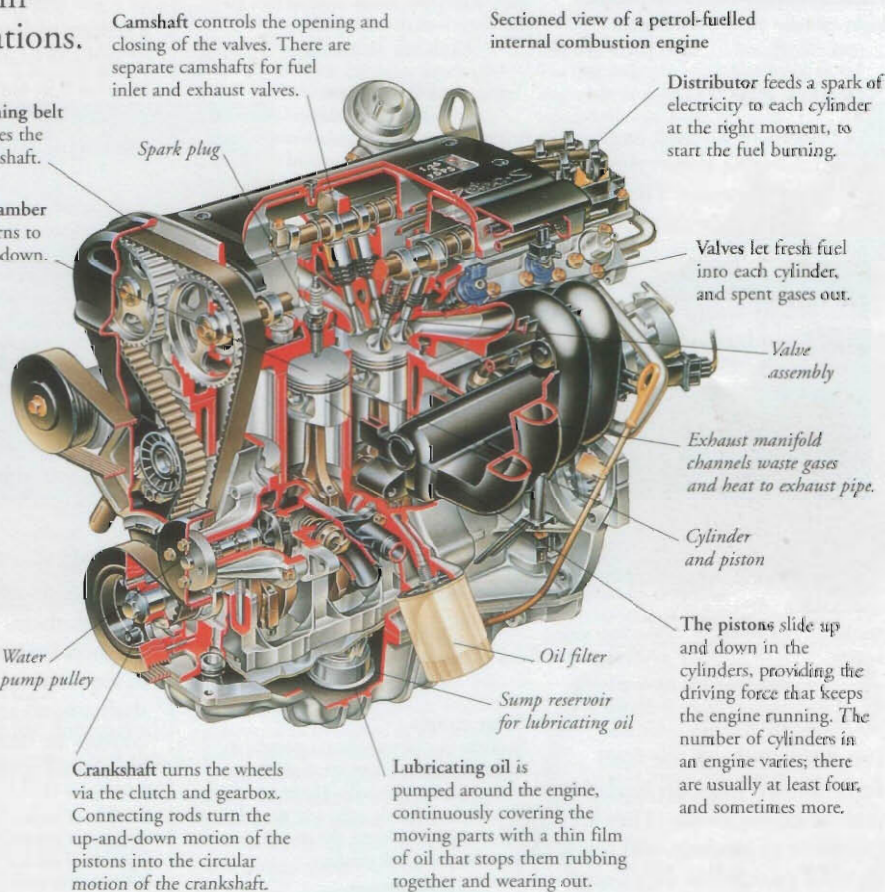
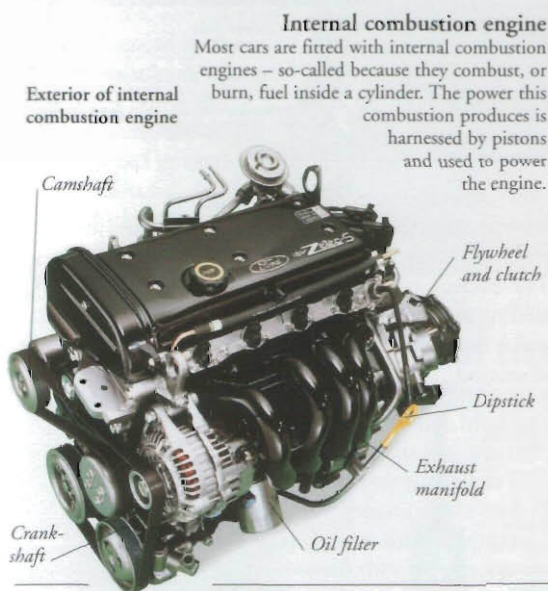
# ENGINES AND MOTORS



EVERY MACHINE THAT MOVES OR HAS moving parts needs an engine or a motor to make it work. A motor is a machine that converts some form of energy, such as fuel or electricity, into motion. An engine is a form of motor. Engines and motors, both huge and tiny, are everywhere – in vehicles from motor cycles to airliners and railway locomotives, and in appliances around the house, in industrial machines, and in power stations.

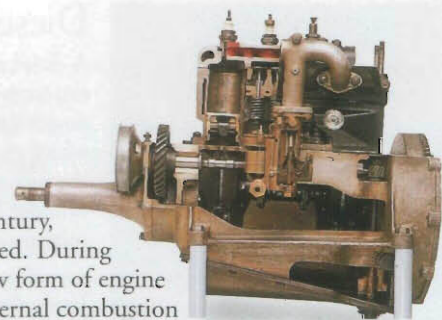
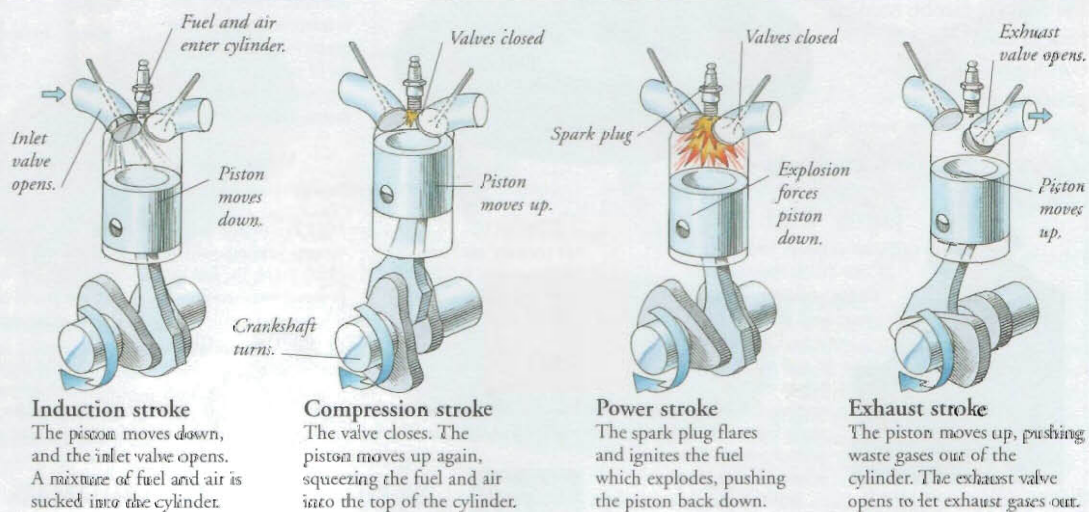
## Modern engines

Fuel efficiency, plenty of power for its low weight, and little need for maintenance are the hallmarks of the modern car engine. Many engines have electronic components that increase their fuel efficiency further.



## How engines work

This sequence of diagrams shows what happens in one cylinder of a petrol engine while the engine is running. During the sequence, the piston goes down, up, down, and up again. This is called a four-stroke cycle. The cycle is repeated over and over again – up to 50 times a second when the engine is turning at high speed. In an engine with more than one cylinder, the cylinders fire one after the other to provide continuous power.



Early four-cylinder petrol engine

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